# Current: charge flow rate

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate current as charge transferred per time.

**SUCCESS CRITERIA - I can...**

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Both charge magnitudes become 3 times as large and separation becomes 5 times as large. Express new force as a fraction of the old.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** I = Δq/Δt.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

**Worked reasoning:** If 12 C pass a point in 4 s, average current is I=Q/t=3 A. Charge circulates; a resistor does not consume charge.

**Your turn:** Draw a complete loop and label equal current before and after one resistor in steady state.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Draw conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?




Circuit symbols: long/short lines = battery; rectangles = resistors.

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A wire carries .25 A for 8 minutes. Find charge transferred.

**Marzano check - circle:** ① ② ③

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### 05 Analyze the reasoning

At a junction, 5 A enters and 2 A leaves along one branch. Find the other outgoing current in steady operation. Explain the conservation rule.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A source approaching a stationary listener at 20 m/s emits 600 Hz. Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$. Find observed frequency.

**Marzano check - circle:** 1 2 3

### 07 Exit check

Does a large electron drift speed have to equal the speed at which a switched circuit responds? Explain qualitatively.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A current rises linearly from 0 to 4 A during 3 s, then stays 4 A for 2 s. Find total charge from the current-time graph.

**Marzano check - circle:** 1 2 3